

# Lecture 5: Applying Consumer Theory (Chapter 5 · §6.5)

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*Applied Microeconomics — Master Notes · by Jakob V. Stangel*

**Required reading:** Friberg, Chapter 5 — Consumer choice (applied) · §6.5 — Income and substitution effects. *Lecturer: L. Butera — deck "Lecture 5: Applying consumer theory."* (This lecture uses the machinery built in Lecture 4 — the optimum where  $MRS = \text{price ratio}$  — and puts it to work.)

**Résumé.** Lecture 4 found the **best bundle for given prices and income**. Lecture 5 *applies* that result three ways: (1) it **derives the demand curve** — change one price, re-solve, repeat, and the optimal quantities trace out demand (closing the loop back to Lecture 2); (2) it **solves the optimum with calculus** (the substitution and Lagrangian methods) to get demand *functions*  $q(p, Y)$ , and introduces the **dual problem** (expenditure minimization); (3) it opens the "black box" of a price change with **income and substitution effects** — the deeper decomposition that explains *inferior* and *Giffen* goods and the puzzle of **labour supply**.

★ **The heart of the lecture:** a price change moves demand for **two** reasons, and separating them explains almost everything. The **substitution effect** (the good is now relatively dearer → switch away, **always** in the "less" direction) and the **income effect** (the price rise makes you effectively poorer → buy less if the good is **normal**, *more* if **inferior**). **Total effect = substitution effect + income effect**. When a strong, positive income effect on an inferior good *overwhelms* the substitution effect, demand slopes **up** — a **Giffen good**. The same tool cracks the wage puzzle: a raise makes you work *more* (substitution) but also lets you afford more leisure (income).

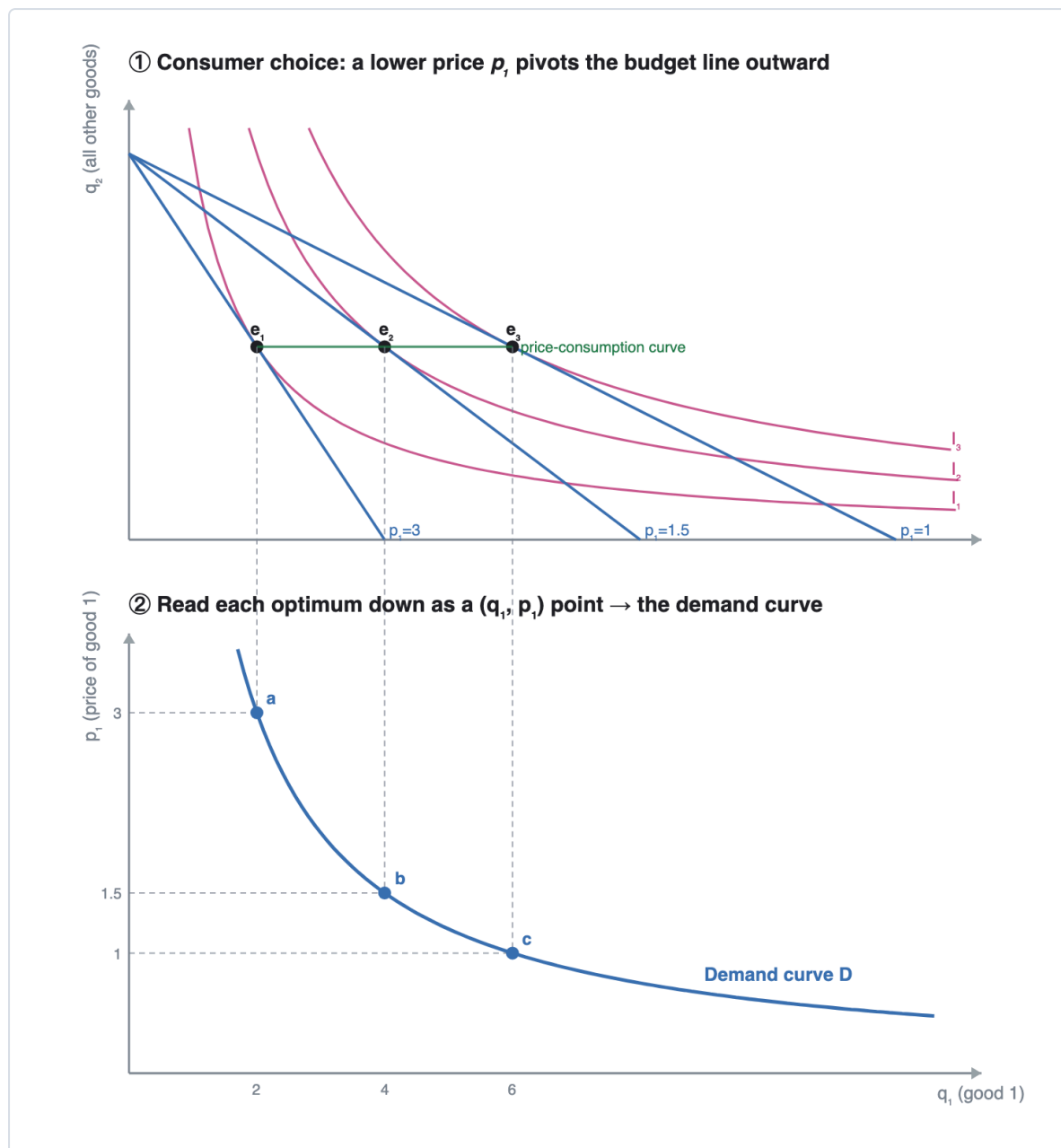
**Key concepts / "modes" to use:** deriving demand from the optimum · the substitution method & the Lagrangian ( $MU_1/p_1 = MU_2/p_2 = \lambda$ ) · demand functions  $q(p_1, p_2, Y)$  · the dual / expenditure minimization / Hicksian (compensated) demand · substitution effect vs income effect (Slutsky/Hicks decomposition) · normal · inferior · Giffen · the labour-leisure model & backward-bending supply.

## ★ Deriving the demand curve from consumer choice

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**Core:** we never really *assumed* the demand curve of Lecture 2 — we can **derive** it. Hold **preferences, income and other prices constant**, then **change the good's own price** and re-solve the optimum. The resulting optimal quantities, plotted against price, **are** the demand curve.

- **The recipe:** at each price  $p_1$ , find the tangency bundle ( $MRS = p_1/p_2$ ). Lower  $p_1$  → the budget line swings out → a **new, higher** optimal  $q_1$ . Collect the ( $p_1$ ,  $q_1$ ) pairs → the **individual demand curve** for good 1.



How to read it (building demand, step by step). ① (top) start from the consumer's optimum. Every time  $p_1$  falls, the budget line **pivots out** along the horizontal axis (same vertical intercept  $Y/p_2$ ), and the tangency slides to a **new optimum** with **more  $q_1$** :  $e_1 \rightarrow e_2 \rightarrow e_3$ . ② (bottom) each optimum hands you one (**quantity, price**) pair — drop a dashed line down from  $e_1$  to plot point  $a = (q_1 \ 2, \ p_1 \ 3)$ , from  $e_2$  to  $b = (4, \ 1.5)$ , from  $e_3$  to  $c = (6, \ 1)$ . **Join a, b, c  $\rightarrow$  the demand curve D**. That's the whole trick: *the demand curve is just the consumer's optimum, re-solved and re-plotted for every price.* (This example is Cobb–Douglas  $U = q_1 q_2$ , so demand is  $q_1^* = Y/(2p_1)$  — the smooth, unit-elastic curve shown.)

- **Movement vs shift (ties back to Ch. 2):** changing the good's **own price** traces a **movement along** its demand curve; changing **income** or **another price** re-solves to a different curve — a **shift of demand**. *In plain terms:* the demand curve is just "the optimum, redrawn for every possible own-price."
- **Why this matters:** it makes demand a *result*, not an assumption — and it means the **shape** of demand (its elasticity, whether it can slope up) is inherited from the **shape of preferences** (the indifference curves).

## Solving the optimum analytically — two methods

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**Core:** the graphical tangency  $MRS = p_1/p_2$  can be solved with **calculus** to get the **demand functions**  $q_1^*(p_1, p_2, Y)$  and  $q_2^*(p_1, p_2, Y)$  explicitly. Two equivalent routes.

### (a) The substitution method

**Core:** turn the constrained problem into an **unconstrained** one by substituting the budget line into utility.

1. **Budget:**  $p_1 q_1 + p_2 q_2 = Y \rightarrow$  solve for one good:  $q_2 = (Y - p_1 q_1)/p_2$ .
2. **Substitute** into  $U(q_1, q_2) \rightarrow$  utility is now a function of  $q_1$  alone.
3. **Maximize:** set  $dU/dq_1 = 0$ . This reproduces the tangency condition  $MU_1/MU_2 = p_1/p_2$  and, with the budget line, yields  $q_1^*$  and  $q_2^*$  as functions of  $(p_1, p_2, Y)$ .

(Requires the utility function to be **quasi-concave** — i.e. indifference curves **convex to the origin** — so the tangency is a maximum, not a minimum.)

### (b) The Lagrangian method

**Core:** the general tool for constrained optimization (met in L4's appendix). Build a function that bolts the constraint onto the objective with a multiplier  $\lambda$ .

$$\mathcal{L} = U(q_1, q_2) + \lambda (Y - p_1 q_1 - p_2 q_2)$$

- **First-order conditions** (set each partial derivative to 0):
  - $\partial \mathcal{L} / \partial q_1 = MU_1 - \lambda p_1 = 0$
  - $\partial \mathcal{L} / \partial q_2 = MU_2 - \lambda p_2 = 0$
  - $\partial \mathcal{L} / \partial \lambda = Y - p_1 q_1 - p_2 q_2 = 0$  (the budget, recovered)
- **Rearrange the first two:**  $MU_1/p_1 = MU_2/p_2 = \lambda$  — the "equal marginal utility per euro" rule: at the optimum the **last euro spent on each good buys the same extra utility**. Dividing them gives back  $MU_1/MU_2 = p_1/p_2$  (MRS = price ratio).
- **What is  $\lambda$ ?** the **marginal utility of income** — how much extra utility one more euro of budget would buy. (A "shadow price" on the budget constraint.)
- Solve the three equations together  $\rightarrow$  the **demand functions**. (Worked for Cobb–Douglas in the practice set — you get the tidy result that a Cobb–Douglas consumer spends **fixed budget shares** on each good.)

## The dual problem — expenditure minimization

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**Core:** utility maximization has a **mirror image**. Instead of "maximise utility for a given budget," ask "**minimise the spending needed to reach a given utility  $\bar{U}$** ." Same tangency, approached from the other side.

$$\min p_1 q_1 + p_2 q_2 \quad \text{subject to} \quad U(q_1, q_2) = \bar{U}$$

- **The solution is the expenditure function  $E(p_1, p_2, \bar{U})$**  — the **least money** needed to achieve utility  $\bar{U}$  at prices  $(p_1, p_2)$ .

- It gives the **Hicksian (compensated) demands** — quantities as functions of prices holding **utility** constant (rather than income). *This is exactly the object the **substitution effect** uses below* — it asks how you'd re-shop if you had to stay on the **same indifference curve** but faced new relative prices.
- **Duality (the payoff)**: max-utility-given-budget and min-budget-given-utility describe the **same tangency point** — two doors into the same room. The primal gives ordinary (Marshallian) demand  $q(p, Y)$  ; the dual gives compensated (Hicksian) demand  $q(p, \bar{U})$  .

## ★ Income and substitution effects (§6.5)

**Core:** the deeper "behind-the-scenes" reading of a price change. When  $p_1$  rises, quantity demanded moves for **two distinct reasons**, and the total is their sum:

$$\text{Total effect} = \text{substitution effect} + \text{income effect}$$

- **Substitution effect** — the change in quantity from the shift in **relative prices alone**, holding utility constant. If  $p_1$  rises, good 1 is now **relatively dearer**, so you switch toward the (now cheaper) other good. **Always negative (or zero)**: a price rise  $\Rightarrow$  substitution away. *(Formally, this is the move along the **original indifference curve** — the Hicksian/compensated response from the dual.)*
- **Income effect** — the change in quantity because the price rise makes you **effectively poorer** (your income now buys less). Sign depends on the good:
  - **normal good**  $\rightarrow$  you buy **less** when poorer  $\Rightarrow$  income effect is **negative** (reinforces substitution);
  - **inferior good**  $\rightarrow$  you buy **more** when poorer  $\Rightarrow$  income effect is **positive** (opposes substitution).

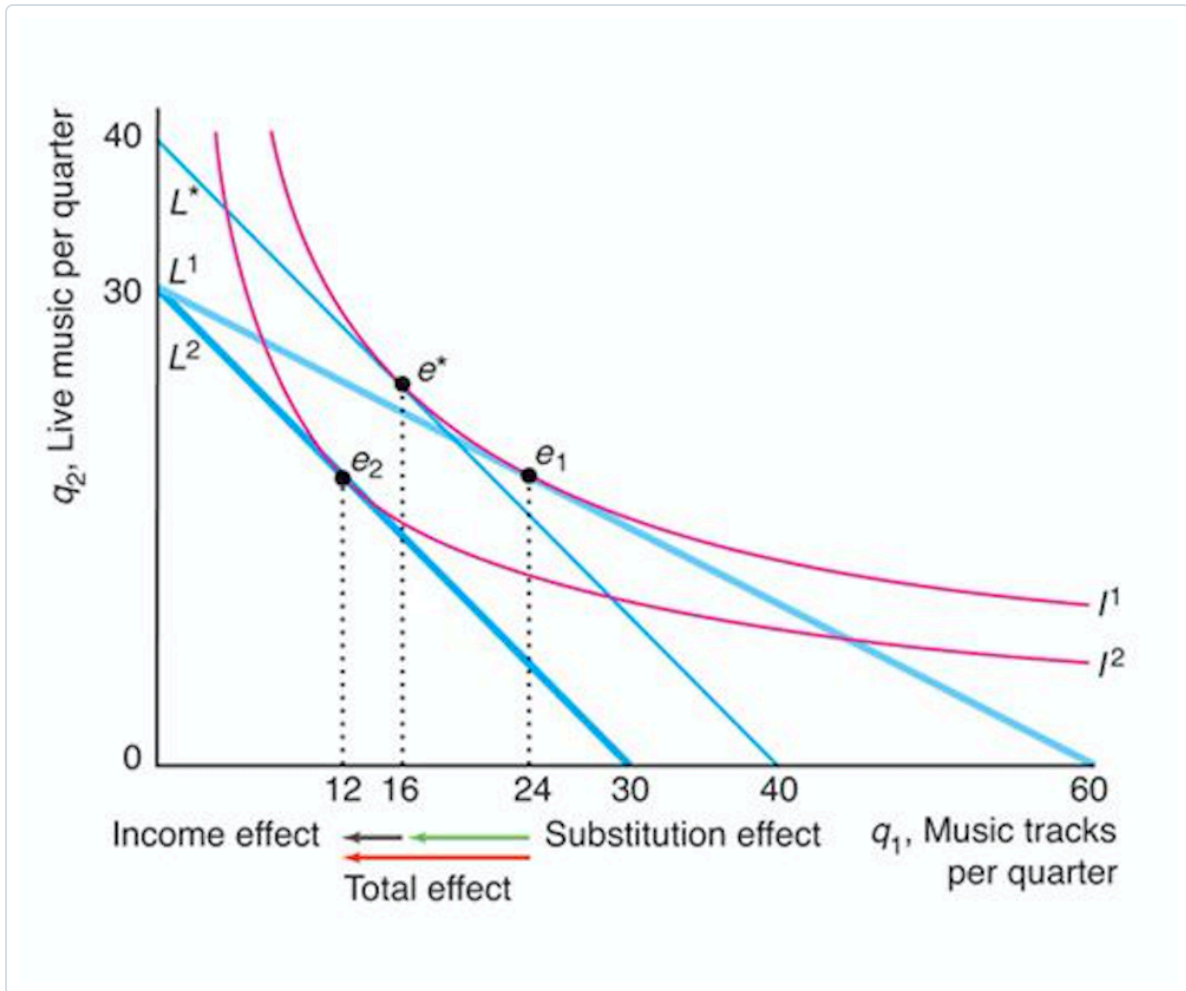
### How the decomposition works (the three-step graph)

**Core:** compensate, then take the compensation away.

1. **Start** at optimum  $e^*$  on indifference curve  $I^1$  with budget line  $L^1$  .
2. **Raise  $p_1$**   $\rightarrow$  budget line pivots inward and **steeper** ( $L^2$ )  $\rightarrow$  new optimum  $e_2$  on a lower curve  $I^2$  .  
The move  $e^* \rightarrow e_2$  is the **total effect**.
3. **Isolate substitution**: slide a budget line with the **new (steeper) slope** out until it is **tangent to the original curve  $I^1$**  (this " $L^*$ " imagines you were **compensated** to stay as happy as before). Its tangency  $e_1$  gives the **substitution effect**  $e^* \rightarrow e_1$  (a slide *along*  $I^1$ ).
4. **Income effect** is the leftover **parallel shift**  $L^* \rightarrow L^2$  , i.e.  $e_1 \rightarrow e_2$  — the pure loss of purchasing power.

### A normal good — the two effects reinforce

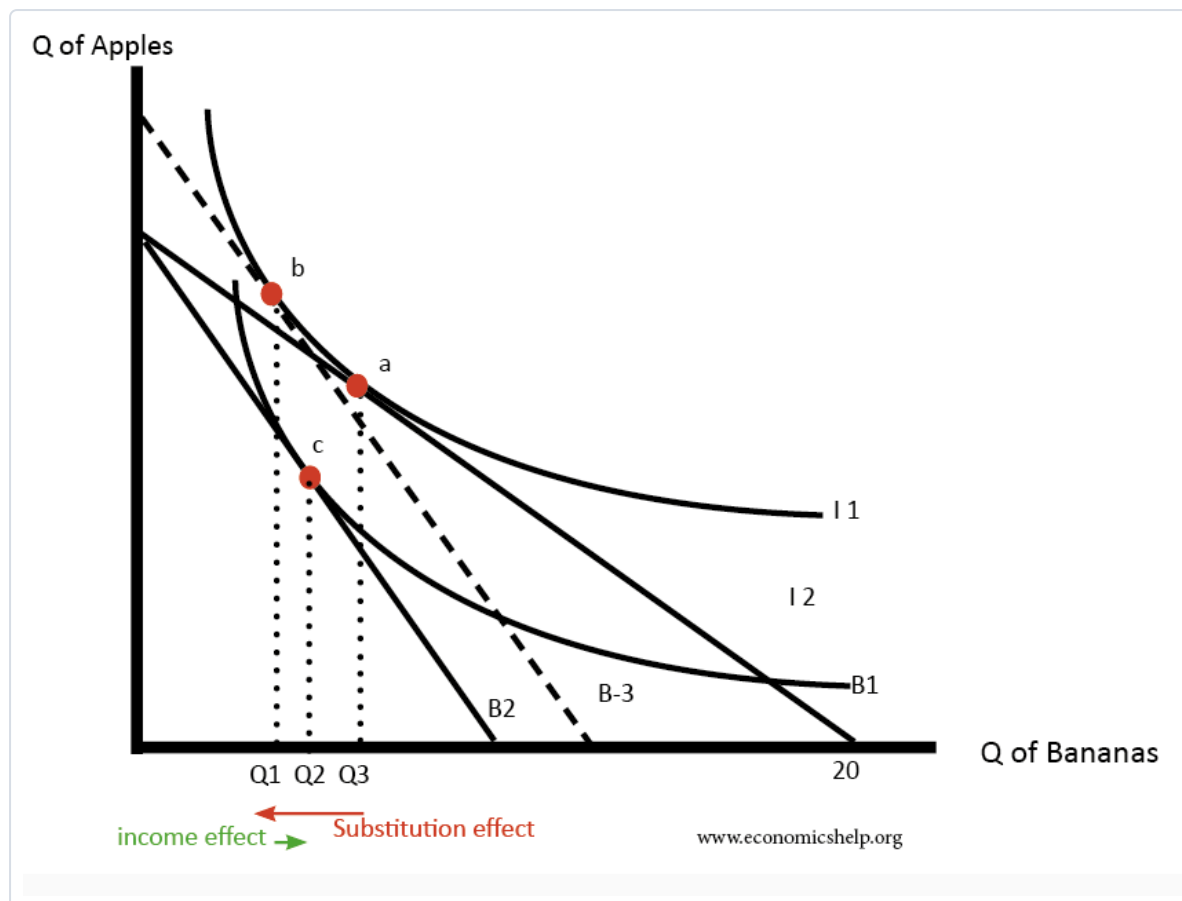
**Core:** for a **normal good** both effects push the **same way**, so demand falls **noticeably** when price rises.



How to read it.  $e^*$  is the original optimum. When the price of music tracks rises, the budget line pivots to  $L^2$  and the optimum falls to  $e_2$ . The compensated line  $L^*$  (new slope, original indifference curve  $I^1$ ) locates  $e_1$ : *substitution effect* =  $e \rightarrow e_1^*$  ( $24 \rightarrow 16$ ), *income effect* =  $e_1 \rightarrow e_2$  ( $16 \rightarrow 12$ ). For a normal good the income arrow points the **same way** as the substitution arrow  $\rightarrow$  the two **add up** to a large total fall ( $24 \rightarrow 12$ ).

## | An inferior good — the effects oppose

**Core:** for an **inferior** good the positive income effect **partly cancels** the substitution effect, so demand is **less responsive** to price.



How to read it. The substitution effect still pushes the good **down** ( $a \rightarrow b$ ), but because the good is **inferior**, being poorer makes you want **more** of it — the income arrow points **back the other way**, offsetting part of the substitution move. **Net**: demand still falls, but by **less** than for a normal good. "The more inferior the good (the more negative its income elasticity) and the fewer the substitutes, the less demand shrinks when price rises."

## **Giffen goods — when demand slopes up**

**Core**: the rare, dramatic case. A **Giffen good** is an **inferior** good whose **income effect** is so **strong** it **dominates** the substitution effect → **quantity demanded rises when price rises** → an **upward-sloping** demand curve, violating the law of demand.

- **The recipe for a Giffen good**: (1) strongly **inferior**; (2) takes up a **large share** of the budget; (3) has **few substitutes**. Then a price rise makes the household so much poorer that it **cuts the "luxury"** it used to afford and buys **even more** of the cheap staple.
- **The classic examples**: **potatoes in Ireland** in the 1840s (the origin of the name — still contested) and, more convincingly, **rice among poor households in China** (Jensen & Miller, 2008): when rice got dearer, families could no longer afford *any* meat, so they bought **more** rice to fill up. → "They can only afford rice."
- **Why it's the exception, not the rule**: it needs a strong income effect (big budget share) *and* strong inferiority — rare in rich economies. The substitution effect usually wins, which is why demand almost always slopes down.

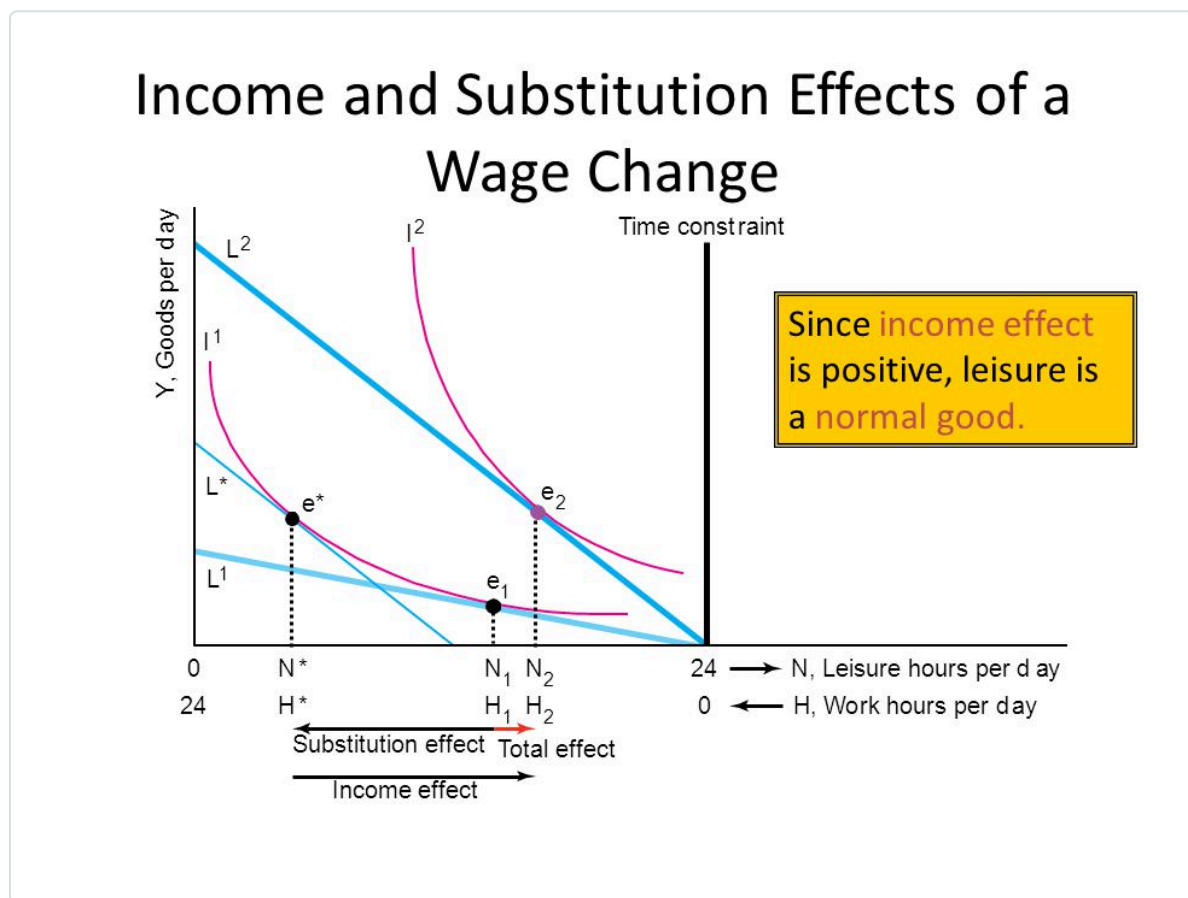
(Aside — **Hicks vs Slutsky**: two ways to "compensate" for the substitution effect. **Hicks** keeps you on the **original indifference curve** (same utility); **Slutsky** keeps the **original basket just affordable** (rotate the budget line around

the starting point). Hicks is the standard version; both give the same intuition.)

## ★ Application: labour supply (the wage puzzle)

**Core:** the decomposition cracks a real puzzle — *does a pay rise make you work more or less?* Model the worker choosing between **leisure** and **consumption (income)**. A wage is the **price of leisure** (an hour off costs you one hour's wage). So a **wage rise = leisure becomes more expensive** → apply income & substitution effects to *leisure*.

- **Substitution effect:** leisure is now dearer → consume less leisure → **work more**.
- **Income effect:** a higher wage makes you **richer**; if leisure is a **normal good**, you want **more** of it → **work less**.
- **Net effect is ambiguous** — it depends on which dominates.



*How to read it.* The horizontal axis is **leisure hours N** (read right-to-left, it's **work hours H**); the vertical axis is **goods/income**. A wage change tilts the budget line. Splitting  $e^* \rightarrow e_2$  at the compensated point  $e_1$ : the **substitution effect** cuts leisure (more work), the **income effect** adds leisure (less work — "since the income effect is positive, leisure is a normal good"). Because they pull **opposite ways**, the **backward-bending labour-supply curve** becomes possible: at **low** wages the substitution effect usually wins (higher wage → work more); at **high** wages the income effect can win (higher wage → work less).

- **The deck's puzzle made concrete:** a raise from 200 → 300 DKK/hr probably makes you **work more** (substitution dominates); a raise from 200 → 4000 DKK/hr might make you **work less** — you're now rich enough to "buy back" your time (income dominates).

## Formula sheet (Lecture 5)

| Tool                                  | Formula                                                                                  |
|---------------------------------------|------------------------------------------------------------------------------------------|
| Tangency (optimum)                    | $MRS = MU_1/MU_2 = p_1/p_2$                                                              |
| Budget constraint                     | $p_1q_1 + p_2q_2 = Y$                                                                    |
| Lagrangian                            | $\mathcal{L} = U(q_1, q_2) + \lambda(Y - p_1q_1 - p_2q_2)$                               |
| Lagrangian FOCs                       | $MU_1 = \lambda p_1, MU_2 = \lambda p_2 \rightarrow MU_1/p_1 = MU_2/p_2 = \lambda$       |
| $\lambda$ interpretation              | <b>marginal utility of income</b> (shadow price of the budget)                           |
| Expenditure function (dual)           | $E(p_1, p_2, \bar{U}) = \text{min spending to reach utility } \bar{U}$                   |
| <b>Slutsky decomposition</b>          | <b>total effect = substitution effect + income effect</b>                                |
| Substitution effect                   | $\leq 0$ <b>always</b> (holding utility constant)                                        |
| Income effect (price $\uparrow$ )     | <b>normal:</b> negative · <b>inferior:</b> positive · <b>Giffen:</b> positive & dominant |
| Cobb–Douglas $U = q_1^a q_2^b$ demand | $q_1^* = a/(a+b) \cdot Y/p_1$ (constant budget shares)                                   |

## ★ Practice problems (with solutions)

Do these — Lecture 5 is where the calculus of consumer choice actually gets tested. Solutions below.

1. A consumer has  $U(q_1, q_2) = q_1 q_2$ , income  $Y$ , prices  $p_1, p_2$ . Use the **substitution method** to find the demand for  $q_1$ .
2. Same consumer, now use the **Lagrangian** method to find  $q_1^*, q_2^*$ , and interpret  $\lambda$ .
3. Show that with **Cobb–Douglas**  $U = q_1^a q_2^b$ , the consumer spends a **constant share** of income on each good. What is the own-price elasticity of the demand  $q_1^* = [a/(a+b)] \cdot (Y/p_1)$ ?
4. The price of good 1 (a **normal** good) rises. On an indifference-curve diagram, identify the **substitution** and **income** effects, and say which direction each pushes  $q_1$ .
5. Classify the income effect (sign, for a **price rise**) and say what happens to total demand: (a) a normal good; (b) an inferior good; (c) a Giffen good.
6. Your hourly wage doubles. Using income & substitution effects, explain why you might **work more or less**. Which effect is the "substitution" one?
7. (Concept.) What does the **expenditure function**  $E(p_1, p_2, \bar{U})$  measure, and how does it relate to the **substitution effect**?

## Solutions

1. Budget:  $q_2 = (Y - p_1 q_1)/p_2$ . Substitute:  $U = q_1 \cdot (Y - p_1 q_1)/p_2$ . Differentiate wrt  $q_1$  and set to 0:  $dU/dq_1 = (Y - 2p_1 q_1)/p_2 = 0 \rightarrow Y - 2p_1 q_1 = 0 \rightarrow$

$$q_1^* = Y / (2p_1)$$

(and by symmetry  $q_2^* = Y/(2p_2)$ ). The consumer spends **half** the budget on each good.

2.  $\mathcal{L} = q_1 q_2 + \lambda(Y - p_1 q_1 - p_2 q_2)$ . FOCs:  $q_2 = \lambda p_1$  and  $q_1 = \lambda p_2$  and the budget. Divide the first two:  $q_2/q_1 = p_1/p_2 \rightarrow p_1 q_1 = p_2 q_2$ . Put into the budget  $p_1 q_1 + p_2 q_2 = Y \rightarrow 2p_1 q_1 = Y \rightarrow q_1^* = Y/(2p_1)$ ,  $q_2^* = Y/(2p_2)$  (same as Q1 ✓).  $\lambda = q_2/p_1 = Y/(2p_1 p_2)$  — the **marginal utility of an extra euro** of income.

3. Lagrangian FOCs give  $MU_1/MU_2 = (a q_1^{a-1} q_2^b)/(b q_1^a q_2^{b-1}) = (a/b)(q_2/q_1) = p_1/p_2 \rightarrow p_1 q_1 = (a/b)p_2 q_2$ . Sub into the budget  $\rightarrow p_1 q_1 = [a/(a+b)] \cdot Y$ . So **spending on good 1 is the fixed share  $a/(a+b)$  of income** (independent of prices)  $\rightarrow q_1^* = [a/(a+b)] \cdot (Y/p_1)$ . **Own-price elasticity** =  $(dq_1/dp_1)(p_1/q_1) = -[a/(a+b)]Y \cdot p_1^{-2} \cdot p_1/q_1 = -1$ : Cobb–Douglas demand is **unit-elastic** (which is exactly why expenditure share is constant — recall [Lecture 3](#): unit elasticity  $\Rightarrow$  spending unchanged).

4. Original optimum  $e^*$  on  $I^1$ . Price  $\uparrow \rightarrow$  new optimum  $e_2$  on a lower  $I^2$  (total effect,  $q_1$  falls). Compensated line (new slope, tangent to  $I^1$ ) gives  $e_1$ : **substitution effect  $e^* \rightarrow e_1$**  pushes  $q_1$  **down** (dearer relative price); **income effect  $e_1 \rightarrow e_2$**  also pushes  $q_1$  **down** (poorer, and it's normal). Both **negative**  $\rightarrow$  they **reinforce**. (*This is the normal-good figure.*)

5. For a **price rise**: (a) **normal**  $\rightarrow$  income effect **negative**; total demand **falls** (both effects down). (b) **inferior**  $\rightarrow$  income effect **positive**, partly offsets substitution; total demand **falls, but less**. (c) **Giffen**  $\rightarrow$  income effect **positive and dominant**; total demand **rises** (law of demand violated).

6. A wage rise makes leisure **more expensive** (its price = the wage). **Substitution effect**: leisure dearer  $\rightarrow$  take less leisure  $\rightarrow$  **work more**. **Income effect**: you're richer and leisure is normal  $\rightarrow$  want more leisure  $\rightarrow$  **work less**. The **substitution effect** is the "leisure-is-now-dearer" one; whether you net work more or less depends on which dominates ( $\rightarrow$  backward-bending supply).

7.  $E(p_1, p_2, \bar{U})$  = the **minimum spending needed to reach utility  $\bar{U}$**  at given prices (the dual). Its quantities are the **Hicksian/compensated demands** — precisely what the **substitution effect** uses, since that effect asks how you re-optimize when **utility is held fixed** and only **relative prices** change.

## ★ Fun facts & memorable details (Lecture 5)

Sticky bits from applying consumer theory.

- **The demand curve is a result, not an assumption.** Wiggle one price, re-solve the optimum over and over, and demand **draws itself**. Its whole shape is inherited from the curvature of your indifference curves.

- **$\lambda$  has a name: the marginal utility of money.** The Lagrange multiplier isn't just algebra — it tells you **how much happier one extra euro would make you**.
  - **Giffen goods really exist — barely.** For 150 years the **Irish-potato** story was economists' only (contested) example. Then **Jensen & Miller (2008)** found it for real: subsidise rice for poor Chinese families and they bought *less* — because they could finally afford some meat.
  - **"They can only afford rice."** The heart of the Giffen paradox: when your staple gets pricier, you're too poor to buy anything *but* the staple — so you buy **more** of it.
  - **A pay rise can make you work less.** The **backward-bending labour supply curve**: past some wage, you're rich enough to "buy back" your own time. Lottery winners who quit their jobs are the income effect made visible.
  - **Every price change is secretly two changes.** Splitting the substitution effect from the income effect is one of the most useful moves in all of microeconomics — it's why we can tell a normal good from an inferior one from a Giffen good just by signing an arrow.
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